

MD KHALEQUZZAMAN CHOWDHURY SAYEM

Current Position: Researcher at UNIST, South Korea

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Research Interests

Vision-Language Models, Multimodal Reasoning, 3D Human-Object Interaction, Continual Learning, Generative Models.

EDUCATION

Ulsan National Institute of Science & Technology

Sep 2023 – August 2025

Masters in Computer Science and Engineering

Ulsan, Korea

Ulsan National Institute of Science & Technology

Sep 2019 – Aug 2023

Bachelor in Industrial Engineering

Ulsan, Korea

RELEVANT PUBLICATIONS AND PROJECT (* denotes equal contribution)

- E. Ismayilzada*, MD K. C. Sayem*, Y. Y. Tiruneh, M. T. Chowdhury, M. Boboev, and S. Baek. QORT-Former: Query-optimized Real-time Transformer for Understanding Two Hands Manipulating Objects. **Accepted at AAAI 2025**. Project Page: <https://kcsayem.github.io/QORT-Former/>
 - As co-first author, I designed the core query optimization method and engineered the integration of contact map features into the query mechanism.
 - Took the lead on writing, synthesizing the technical contributions from all co-authors into the final manuscript for submission.
 - Our framework achieved state-of-the-art accuracy, outperforming prior methods by up to +27.2% on the H2O dataset and +10.4% on the FPFA dataset, while operating at 53.5 FPS (doubling the inference speed).
- **HandVQA**: Diagnosing Fine-Grained Spatial Reasoning Failures in Vision-Language Models via Hand Pose Question Answering. (Masters Thesis/Under review) [Online Version]
 - Proposed the core concept of HandVQA: building upon a pipeline for text generation and transforming it into a multiple-choice VQA format for rigorous, scalable VLM evaluation.
 - Authored the Master's Thesis that introduced the resulting benchmark of over 1.6 million diagnostic questions.
 - Personally ran VLM benchmarking experiments and led the analysis, which demonstrated that base models like LLaVA and Qwen2.5VL exhibit systematic biases and perform at or below random guessing on spatial reasoning tasks.
- MD K. C. Sayem, E. Fozilov, M. Chowdhury, "4D Instruct-GS2G: Editing 4D Gaussians using text instructions". The project was conducted during the 3D Computer Vision Class at UNIST.
 - Implemented the integration of text-derived editing using *InstructPix2Pix* for the project, which developed a method to extend semantic editing to dynamic 3D scenes using text instructions.
 - Achieved realistic, globally consistent edits on real-world scenes with fast training times (30 mins per scene) and high rendering speeds (up to 82 FPS).

RELEVANT WORK EXPERIENCES

Vision & Learning Lab 

Jan 2023 – Present

Researcher (previously Research Assistant during Master's)

UNIST, Ulsan, Korea

- Conducted research in 3D computer vision and vision-language models, resulting in a co-first author publication at AAAI 2025 and a Master's thesis.
- Responsible for designing novel Transformer architectures, conducting large-scale experiments, and preparing manuscripts for top-tier academic conferences.
- Collaborated with a team on developing and benchmarking real-time models for 3D human-object interaction.

Machine Learning & Vision Lab

Sep 2021 – Feb 2022

Research Assistant

UNIST, Ulsan, Korea

- Investigated and implemented advanced data augmentation techniques to improve model generalization on the COCO benchmark for 2D Human Pose Estimation.

Statistical Decision Making Lab

Dec 2020 – Aug 2021

Research Assistant

UNIST, Ulsan, Korea

- Reduced computational cost of a news article recommendation system by parallelizing **bandit algorithms**.
- Achieved significant acceleration (up to 3.48x) in parallelized versions of LinUCB and Thompson Sampling.

TECHNICAL SKILLS

Language/Frameworks: Python, Numpy, Scikit-learn, Pandas, OpenCV, Tensorflow, Pytorch, Networkx, Linux, Git

TEACHING ASSISTANT EXPERIENCES

Introduction to AI Programming in Python

- **Instructor:** Professor Michael Jon Burrell
- **Semesters:** Fall 24, Spring 24, Fall 23

Discrete Mathematics

- **Instructor:** Professor Michael Jon Burrell
- **Semester:** Fall 24

Advanced Computer Vision

- **Instructor:** Professor Seungryul Baek
- **Semester:** Spring 24

LEADERSHIP

PAIS- Partnership for AI and Society

Jan 2021 – Present

President

UNIST

- Initiated a club with the help of **Prof. Bradley Tatar** & students associated with AI research or interested in AI to discuss the risk of AI in our society and how can it be prevented.
- Organized forum and talks to improve awareness on *Data Privacy* among UNIST students.

HONORS & AWARDS

Graduate School Scholarship

Funded By: Korean Government

- The scholarship covers the tuition fees, meal plan & monthly stipends for two years of masters.

UNIST Dream Scholarship

Issued By: UNIST

- The scholarship covers the tuition fees, meal plan & monthly stipends for four years of bachelor.

Acquired \$5K funding through AI Challengers Program(AICP)

Issued By: UNIST Innovative Education Center

- The award is given upon the completion of the project “Parallelization of bandit algorithms to reduce the computational cost of news article recommendation system.”

REFERENCES

Professor Seungryul Baek

Supervisor

Masters

- **Personal website:** <https://sites.google.com/site/bsrvision00/>
- **Affiliation:** Associate Professor, Artificial Intelligent Graduate School (AIGS), UNIST, South Korea
- **Affiliation:** Associate Professor, Department of Computer Science and Engineering, UNIST, South Korea
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Professor Binod Bhattarai

Co-supervisor

Masters

- **Personal website:** <https://sites.google.com/view/bbinod/home>
- **Affiliation:** Lecturer, University of Aberdeen, Aberdeen, UK
- **Affiliation:** Honorary Lecturer, University College London, UK
- **Email:** binod.bhattarai@abdn.ac.uk